Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.6 – Electrical Power

Unit 2: Electricity & Energy

The question: *a bulb and a kettle both run on electricity — so why does the kettle use so much more of it, so much faster?* The answer is **power**: how fast energy is transferred.

Learning intentions

All students will state the power formula and use it to calculate **power**.
Most students will use a second power formula, built from resistance.
Some students will calculate how much **energy** (and so how much it costs) to run something for a given time.

Part 1 — Power

Power is how much energy is transferred *every second*. It is measured in **watts (W)**, and it is linked to voltage and current by:

$$P = V \times I$$

1) **Calculate** the power of a bulb, if the voltage across it is **6 V** and the current through it is **0.5 A**.

Build it

Build a circuit with one bulb, an ammeter and a voltmeter. Record both readings and calculate the power.

$V =$ _____ V $I =$ _____ A $P = V \times I =$ _____ W

Check: if your bulb has a power rating printed on it, compare your answer to it — they should be close.

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Part 2 — A second way to find power

Last lesson's formula was $V = I \times R$. Substitute that straight into $P = V \times I$, in place of V:

$$P = V \times I = (I \times R) \times I = I^2 \times R$$

$$P = I^2 \times R$$

Same formula, same answer — just useful when you know the current and the resistance, but not the voltage.

2) **Calculate** the power of a **10 Ω** heating element, if a current of **2 A** flows through it.

Mixed practice

Calculate the power in each case. Show which formula you used.

Given	Power
$V = 12\,\text{V}$, $I = 3\,\text{A}$	
$I = 1.5\,\text{A}$, $R = 8\,\Omega$	
$V = 230\,\text{V}$, $I = 4\,\text{A}$	

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Some students — energy and cost

Power is energy *per second*, so over a longer time:

energy transferred = power × time

The more watts, and the longer it runs, the more energy it uses — and the more it costs.

3) **Calculate** how much more energy a **2000 W** kettle transfers in one minute than a **60 W** bulb left on for the same minute. Give both energies in joules (energy in joules = power in watts × time in seconds).

Part 3 — On your own

A string of **LED** fairy lights is rated at **24 W**. An older string of filament fairy lights, the same length, draws **1.2 A** from the **230 V** mains.

4) **Calculate** the power of the filament string, and state which string of lights is cheaper to leave on overnight, and by roughly how many times.

HOMEWORK

Find the power rating (in W) on one appliance at home — it is often printed on a label or in the instructions. Write down the appliance and its power, then calculate how much energy (in joules) it would transfer if left running for **10 minutes** (600 seconds).